Hyperbola: Formula Sheet

Recognition and definition

Hyperbola: $\Delta \neq 0,\ h^2 > ab$

Rectangular: $a + b = 0$

Definition: $|PS_1 - PS_2| = 2a$

Focus–directrix: $PS = e\,PM,\ e > 1$

Standard hyperbola

Equation:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Eccentricity:

$$e^2 = 1 + \frac{b^2}{a^2}$$

Relation: $b^2 = a^2(e^2 - 1)$

Foci: $(\pm ae, 0)$

Directrices:

$$x = \pm\frac{a}{e}$$

LR ends:

$$\left(\pm ae, \pm\frac{b^2}{a}\right)$$

Length of LR:

$$\frac{2b^2}{a} = 2a(e^2 - 1)$$

Vertical: $\dfrac{y^2}{a^2} - \dfrac{x^2}{b^2} = 1$, foci $(0, \pm ae)$

Focal distances, parametric form

Focal distances: $|ex_1 \mp a|$

Difference: $|PS_1 - PS_2| = 2a$

Product: $PS_1 \cdot PS_2 = CP^2 - a^2 + b^2$

Auxiliary circle: $x^2 + y^2 = a^2$

Point: $(a\sec\theta,\ b\tan\theta)$

Alternate definition:

$$\frac{p_2^2}{a^2} - \frac{p_1^2}{b^2} = 1$$

Conjugate and rectangular

Conjugate:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$

Eccentricities:

$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$$

Rectangular: $x^2 - y^2 = a^2,\ e = \sqrt{2}$

Rectangular: $\ell_{LR} = \ell_{TA} = \ell_{CA} = 2a$

Point, chord, line

Position: $S_1 > 0$ in, $S_1 < 0$ out

Chord PQ:

$$\frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}$$

Focal chord:

$$\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-e}{1+e}$$

Tangency: $c^2 = a^2m^2 - b^2$

Director circle: $x^2 + y^2 = a^2 - b^2$

Tangent and normal

Tangent at point:

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Tangent at θ:

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$$

Slope form: $y = mx \pm \sqrt{a^2m^2 - b^2}$

Point of contact:

$$\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$$

Normal at point:

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$$

Normal at θ:

$$ax\cos\theta + by\cot\theta = a^2 + b^2$$

Normality:

$$c^2 = \frac{m^2(a^2+b^2)^2}{a^2 - b^2m^2}$$

Tangents at α, β meet:

$$\left(\frac{a\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}}, \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha+\beta}{2}}\right)$$

Chords and highlights

Mid-point chord: $T = S_1$

Chord of contact / polar: $T = 0$

Pair of tangents: $SS_1 = T^2$

Pole of $lx + my + n = 0$:

$$\left(-\frac{a^2l}{n}, \frac{b^2m}{n}\right)$$

Diameter:

$$y = \frac{b^2}{a^2m}x$$

Perpendiculars from foci: $p_1p_2 = b^2$

Feet from focus: $x^2 + y^2 = a^2$

Asymptotes

Pair:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Relation: $H + C = 2A$

Angle:

$$\tan\theta = \frac{2ab}{a^2 - b^2}$$

Eccentricity:

$$e = \sec\frac{\theta}{2}$$

Triangle with tangent: area $= ab$

Rectangular hyperbola $xy = c^2$

Point:

$$\left(ct, \frac{c}{t}\right)$$

Foci: $\pm(\sqrt{2}c, \sqrt{2}c)$

Tangent at t: $x + t^2 y = 2ct$

Normal at t: $t^3 x - ty = c(t^4 - 1)$

Normal meets again: $t^3 t_1 = -1$

Chord $t_1 t_2$: $x + t_1 t_2 y = c(t_1 + t_2)$

Mid-point chord:

$$\frac{x}{h} + \frac{y}{k} = 2$$

Circle meets: $t_1 t_2 t_3 t_4 = 1$